MAPÚA UNIVERSITY

Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

Names:	Molar, Jabez C. Oabel, Edmarc Justin C.	**Date:**	August 30, 2026
Program:	BS CPE		

A. Learning Outcomes:

At the end of this laboratory, students should be able to:

1. Differentiate Strings, Lists, Tuples, and Dictionaries.
2. Apply appropriate Python data structures in software design.
3. Develop a simple console-based application using collections.
4. Demonstrate teamwork through collaborative software development.
5. Analyze the advantages and limitations of different data structures.

B. Introduction:

Python provides several built-in data structures used to organize and manage data efficiently.
1. Strings
Strings are used to store text data such as:

```
name = "Juan Dela Cruz"
course = "BS Computer Engineering"
```

Use Strings For:

- Student Name
- Course
- Labels and Output Messages

2. Lists
Lists are used to store multiple subjects.

```
subjects = [
"Programming",
"Calculus",
"Physics"
]
```

Use Lists For:

- Subject Management
- Displaying Subject Information

3. Tuples
Tuples store fixed information that should not be changed.

```
student_id = (2026, 1001)
```

Use Tuples For:

- Student ID
- Year and Student Number

4. Dictionary
Store complete record:

```
student = {
"Name": name,
"Course": course,
"Student ID": student_id,
"Subjects": subjects
}
```

Used as the primary data container.

Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

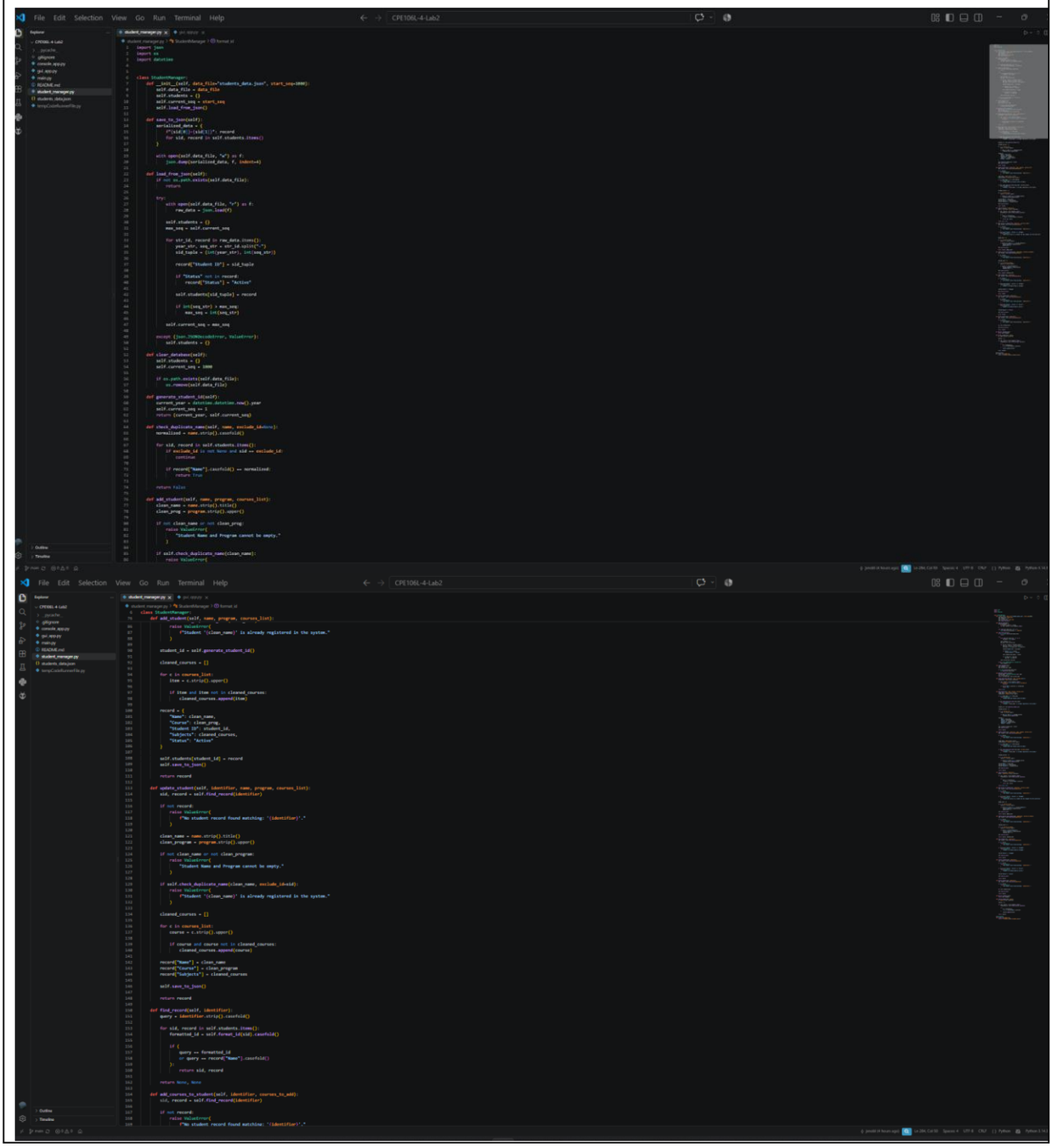
C. Laboratory Scenario:

A university needs a simple **Student Information Management System**. 2 Members
The system should:

1. Store student names.
2. Manage enrolled subjects.
3. Generate a unique student identifier.
4. Save student details using dictionaries.

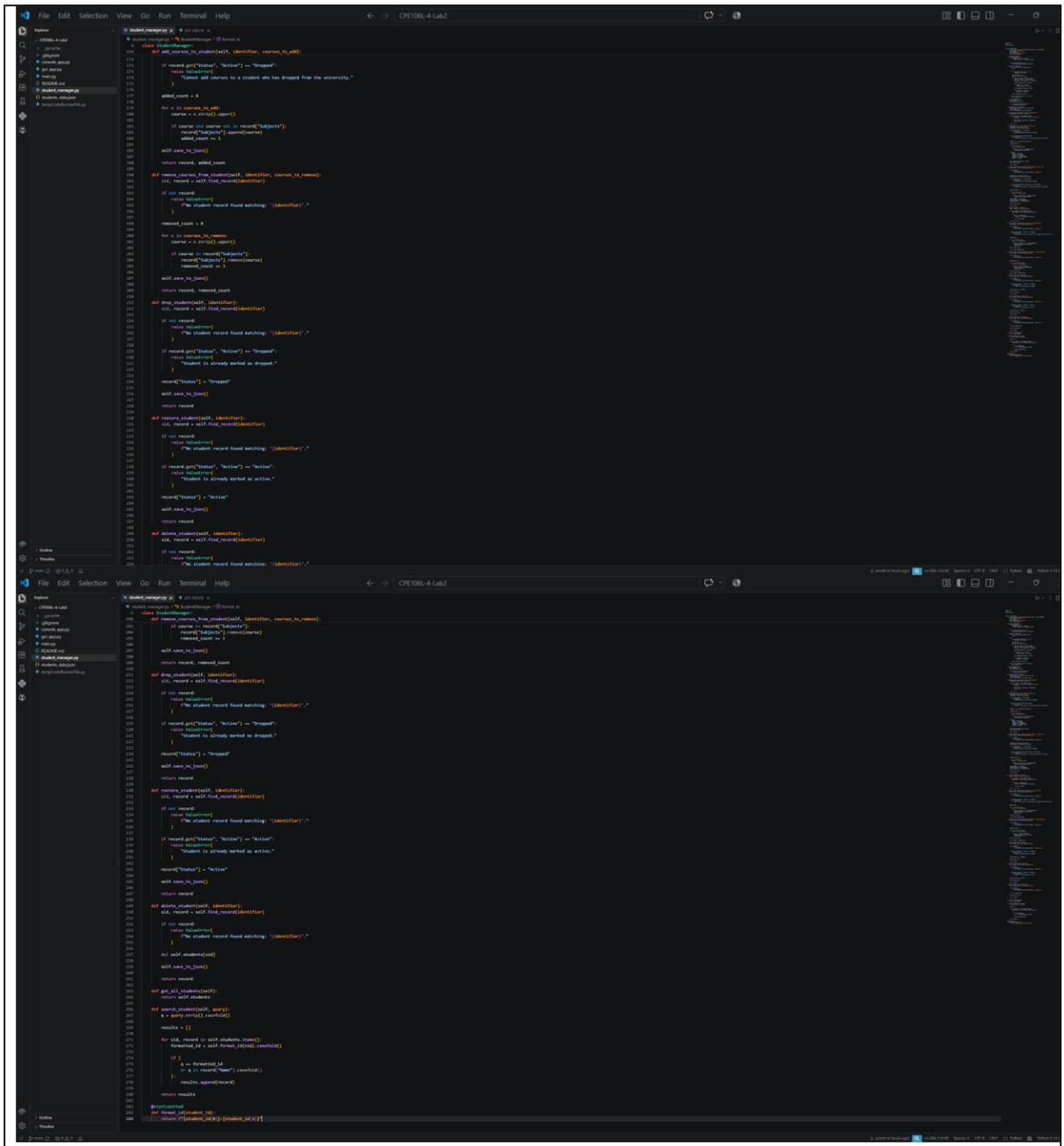
(Please include the following: **1.** Screenshots of the code **2.** Screenshots of the output)

Screenshots of Code (Name of Members) – MOLAR, OABEL
student_manager.py



Laboratory Report 2

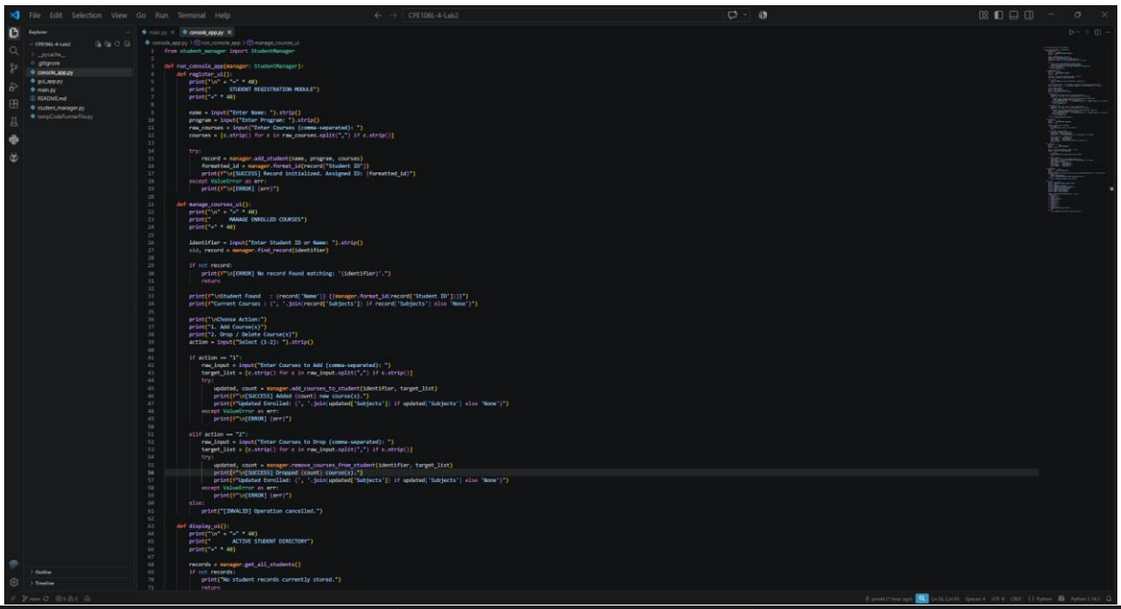
Strings, Lists, Tuples, and Dictionaries

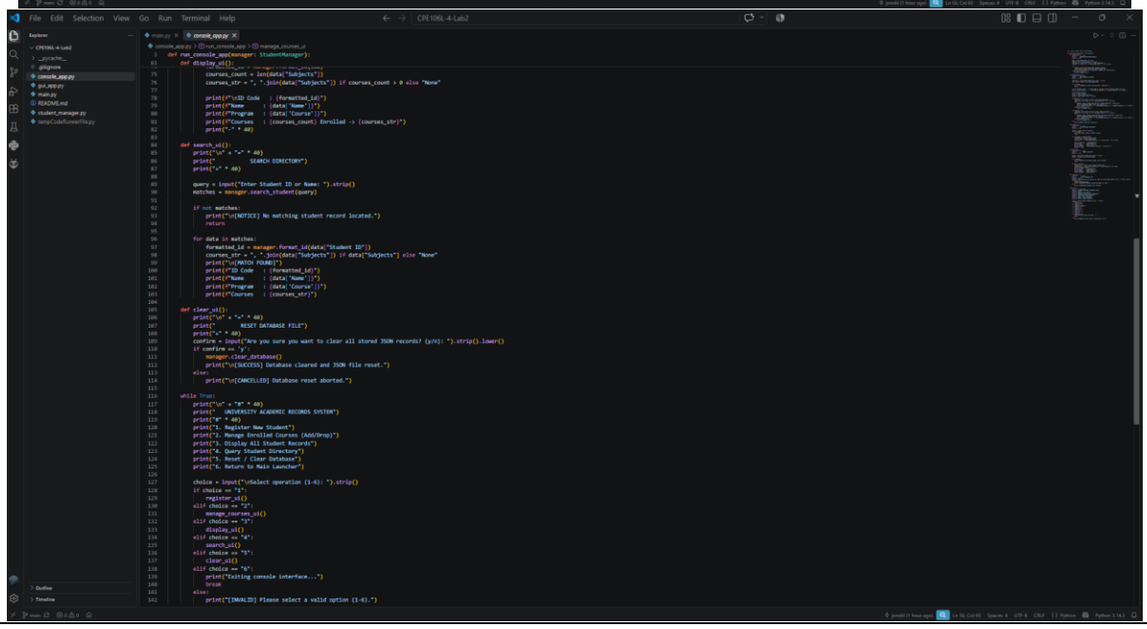


Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

console_app.py





Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

Screenshots of Output

```
========================================
  CPE106L - APPLICATION ENTRY LAUNCHER
========================================
1. Launch Console-Based System
2. Launch Graphical User Interface (GUI)
3. Reset All JSON Data
4. Exit Application

Enter choice (1-4): 1

########################################
   UNIVERSITY ACADEMIC RECORDS SYSTEM
########################################
1. Register New Student
2. Manage Enrolled Courses (Add/Drop)
3. Display All Student Records
4. Query Student Directory
5. Reset / Clear Database
6. Return to Main Launcher

Select operation (1-6): 3

======================================
       ACTIVE STUDENT DIRECTORY
======================================

ID Code   : 2026-1001
Name      : Edmarc Justin C. Oabel
Program   : BS COMPUTER ENGINEERING
Courses   : 7 Enrolled -> CPE106L-4, ECEA101-1, CPE117-4, CPE106-4, CPE105-4, ECEA101-1L, IE103-1
--------------------------------------

ID Code   : 2026-1002
Name      : Jabez C. Molar
Program   : BS COMPUTER ENGINEERING
Courses   : 7 Enrolled -> CPE106L-4, ECEA101-1, ECEA101-1L, CPE106-4, CPE105-4, CPE121-4, CPE117-4
--------------------------------------

########################################
   UNIVERSITY ACADEMIC RECORDS SYSTEM
########################################
1. Register New Student
2. Manage Enrolled Courses (Add/Drop)
3. Display All Student Records
4. Query Student Directory
5. Reset / Clear Database
6. Return to Main Launcher

Select operation (1-6): 1
```

Strings, Lists, Tuples, and Dictionaries

```
[CANCELLED] Database reset aborted.

########################################
   UNIVERSITY ACADEMIC RECORDS SYSTEM
########################################
1. Register New Student
2. Manage Enrolled Courses (Add/Drop)
3. Display All Student Records
4. Query Student Directory
5. Reset / Clear Database
6. Return to Main Launcher

Select operation (1-6): 2

======================================
      MANAGE ENROLLED COURSES
======================================
Enter Student ID or Name: Justin De Leon

Student Found   : Justin De Leon (2026-1003)
Current Courses : CPE106L-4, ECEA101-1, CPE117-4, CPE106-4, CPE105-4, ECEA101-1L, IE103-1

Choose Action:
1. Add Course(s)
2. Drop / Delete Course(s)
Select (1-2): 2
Enter Courses to Drop (comma-separated): IE103-1

[SUCCESS] Dropped 1 course(s).
Updated Enrolled: CPE106L-4, ECEA101-1, CPE117-4, CPE106-4, CPE105-4, ECEA101-1L

########################################
   UNIVERSITY ACADEMIC RECORDS SYSTEM
########################################
1. Register New Student
2. Manage Enrolled Courses (Add/Drop)
3. Display All Student Records
4. Query Student Directory
5. Reset / Clear Database
6. Return to Main Launcher

Select operation (1-6): 3

======================================
       ACTIVE STUDENT DIRECTORY
======================================

ID Code    : 2026-1001
```

Strings, Lists, Tuples, and Dictionaries

```
Name      : Edmarc Justin C. Oabel
Program   : BS COMPUTER ENGINEERING
Courses   : 7 Enrolled -> CPE106L-4, ECEA101-1, CPE117-4, CPE106-4, CPE105-4, ECEA101-1L, IE103-1
---------------------------------------

ID Code   : 2026-1002
Name      : Jabez C. Molar
Program   : BS COMPUTER ENGINEERING
Courses   : 7 Enrolled -> CPE106L-4, ECEA101-1, ECEA101-1L, CPE106-4, CPE105-4, CPE121-4, CPE117-4
---------------------------------------

ID Code   : 2026-1003
Name      : Justin De Leon
Program   : BS COMPUTER ENGINEERING
Courses   : 6 Enrolled -> CPE106L-4, ECEA101-1, CPE117-4, CPE106-4, CPE105-4, ECEA101-1L
---------------------------------------

########################################
   UNIVERSITY ACADEMIC RECORDS SYSTEM
########################################
1. Register New Student
2. Manage Enrolled Courses (Add/Drop)
3. Display All Student Records
4. Query Student Directory
5. Reset / Clear Database
6. Return to Main Launcher

Select operation (1-6): 6
Exiting console interface...

========================================
  CPE106L - APPLICATION ENTRY LAUNCHER
========================================
1. Launch Console-Based System
2. Launch Graphical User Interface (GUI)
3. Reset All JSON Data
4. Exit Application

Enter choice (1-4):
```

Laboratory Report 2

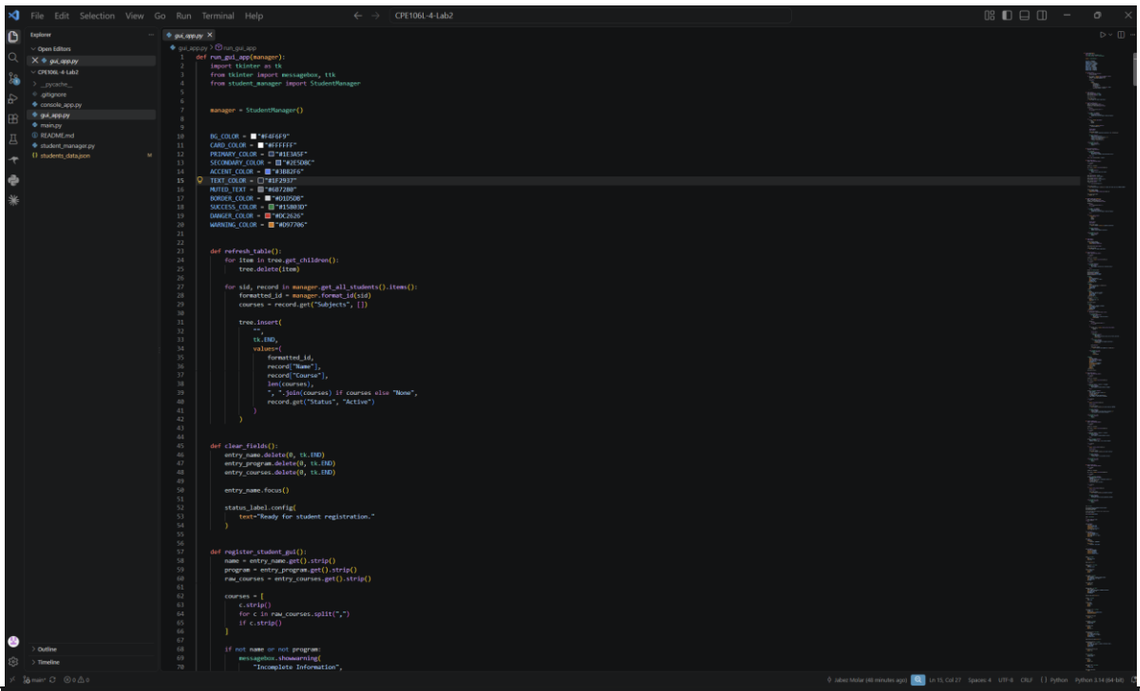
Strings, Lists, Tuples, and Dictionaries

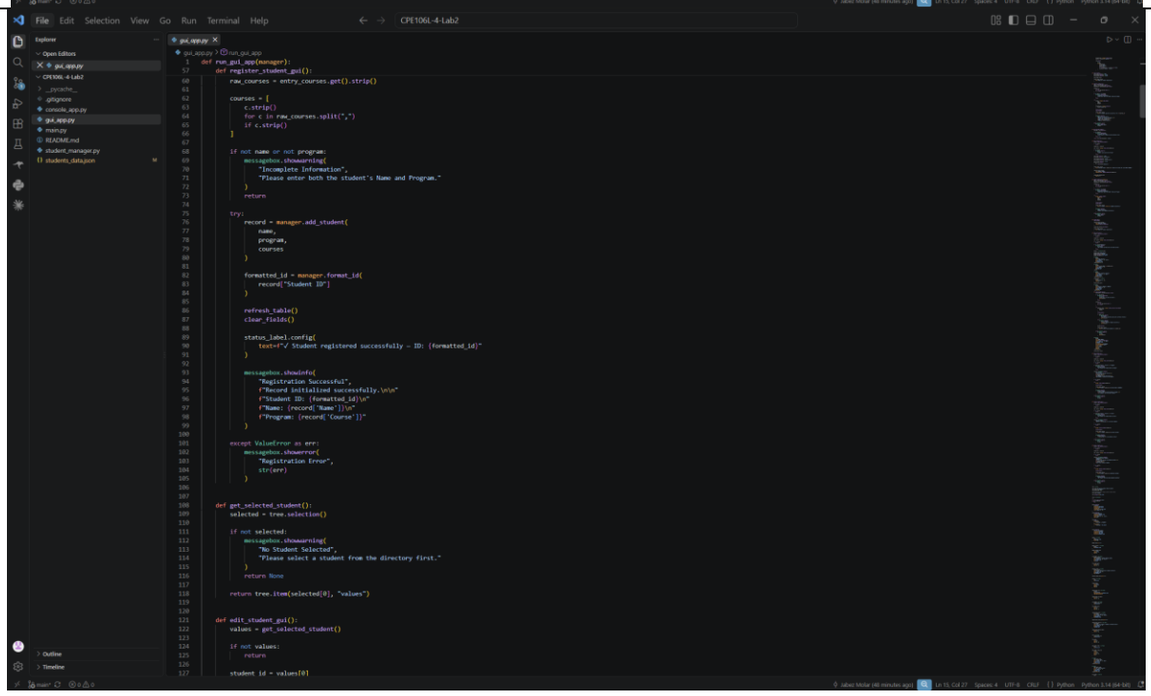
5. Basic GUI using Tkinter. No need to be creative yet, just create a functional GUI and embed the system. 2 Members

(Please include the following: **1.** Screenshots of the code **2.** Screenshots of the output)

Screenshots of Code (Name of Members) - MOLAR, OABEL

gui_app.py





main.py

MAPÚA UNIVERSITY

Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

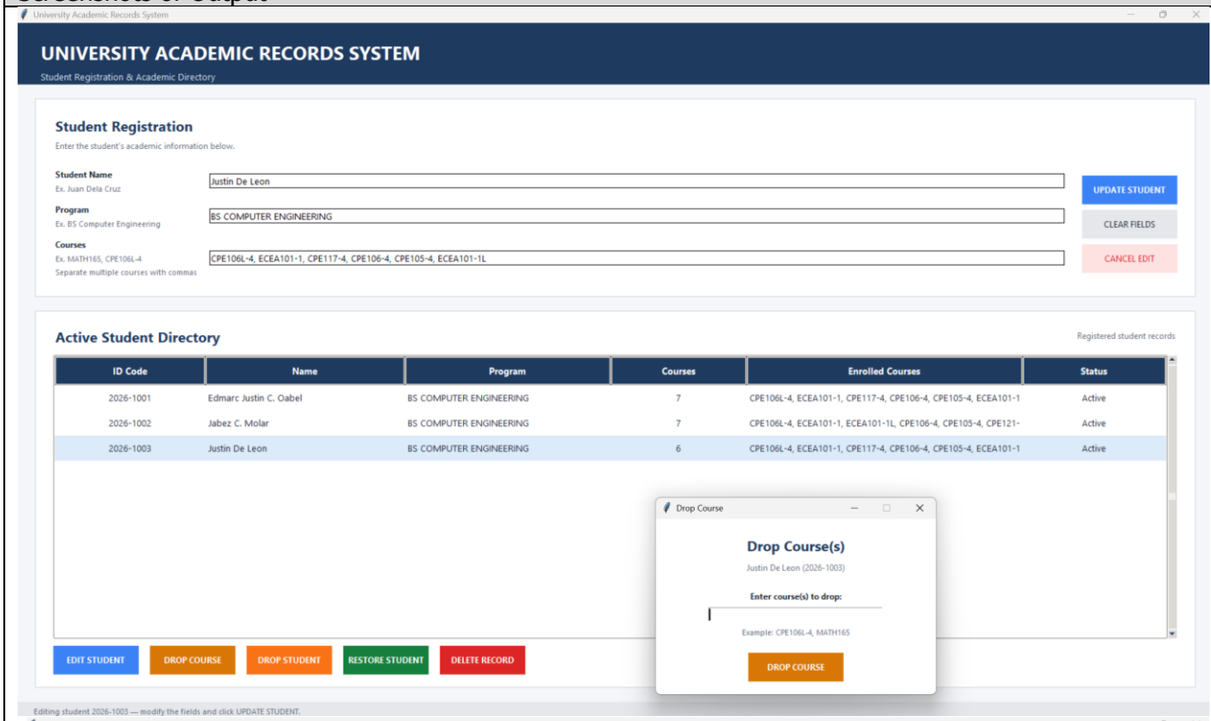
main.py

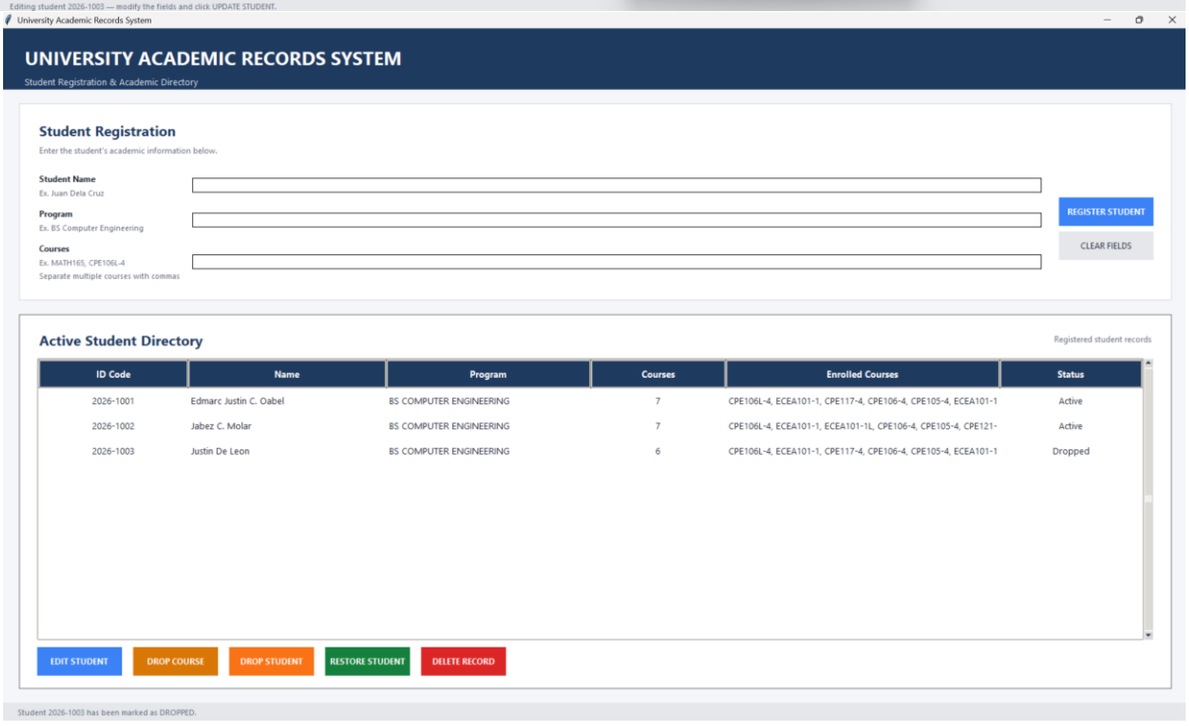
```python
import sys
from student_manager import StudentManager
from console_app import run_console_app
from gui_app import run_gui_app

def main():
    manager = StudentManager()

    while True:
        print("\n========================================")
        print("  CPE106L - APPLICATION ENTRY LAUNCHER   ")
        print("========================================")
        print("1. Launch Console-Based System")
        print("2. Launch Graphical User Interface (GUI)")
        print("3. Reset All JSON Data")
        print("4. Exit Application")

        choice = input("\nEnter choice (1-4): ").strip()

        if choice == "1":
            run_console_app(manager)
        elif choice == "2":
            run_gui_app(manager)
        elif choice == "3":
            confirm = input("Reset stored data file? (y/n): ").strip().lower()
            if confirm == 'y':
                manager.clear_database()
                print("JSON storage file reset successfully.")
        elif choice == "4":
            print("\nShutting down application. Goodbye!")
            sys.exit(0)
        else:
            print("\nInvalid choice. Please input 1, 2, 3, or 4.")

if __name__ == "__main__":
    main()
```

Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

Screenshots of Output





Strings, Lists, Tuples, and Dictionaries

D. Findings, Observations, and Comments:

Guide Questions:

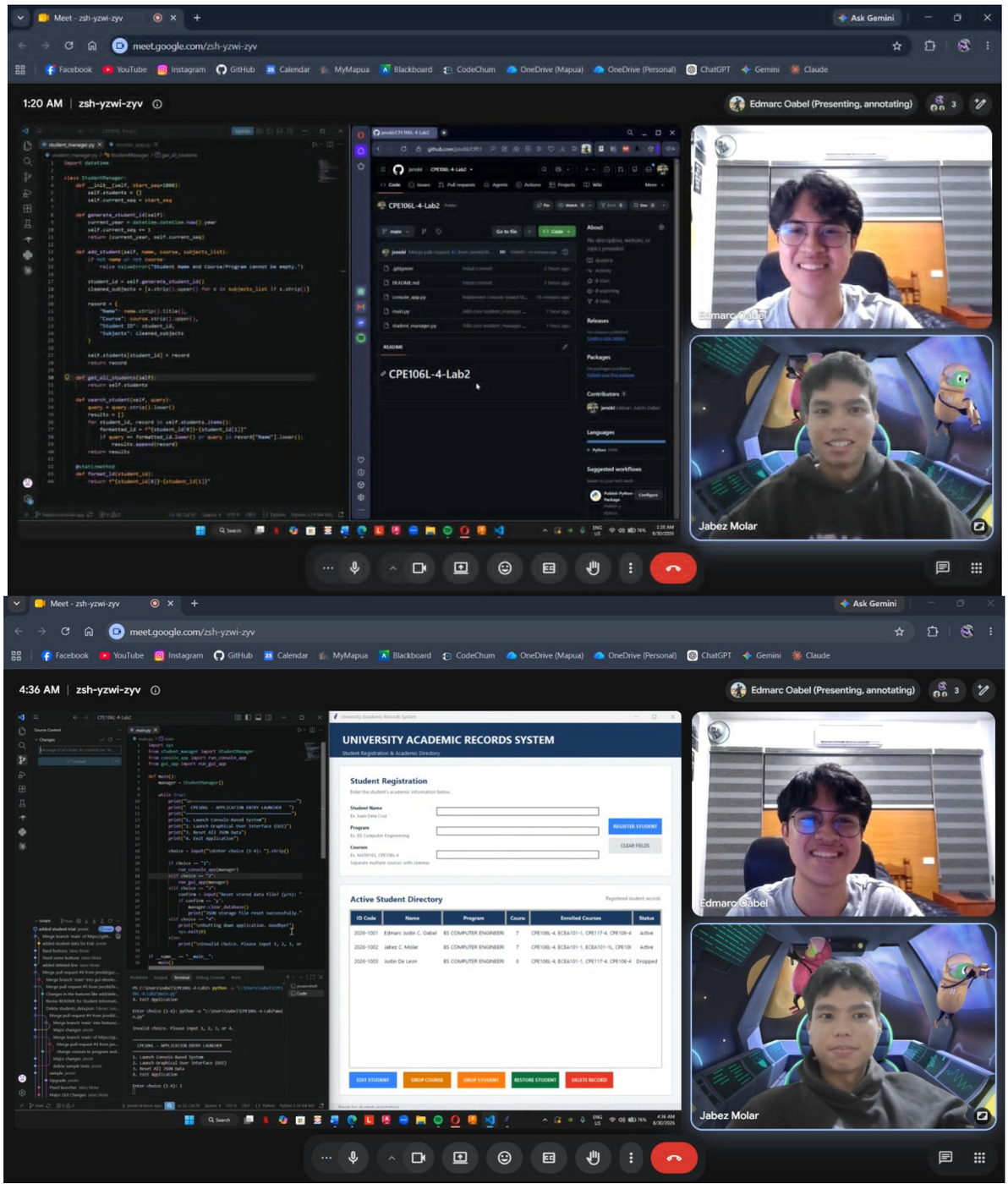
1. What are your Group's realizations in this laboratory?
2.

Through this activity, our group realized how GitHub and VS Code can be used for collaboration and software development. We learned how to use push, pull, merge, and branches to manage our code and work together without affecting each other's work. The activity also taught us the purpose of these features and how they can help organize our projects, track changes, and avoid conflicts. We also realized the importance of communication when working with groupmates, especially when deciding who would work on specific parts of the project. Using branches allowed us to work on our assigned tasks separately and combine our work when it was ready.

Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

PICTURE/PROOF OF DOING THE ACTIVITY



Laboratory Report 2

Strings, Lists, Tuples, and Dictionaries

E. Score Sheet

Criteria	Poor (25%)	Fair (50%)	Good (75%)	Excellent (100%)	Score
I. Completeness, Documentation, and Organization of Report	The report was incomplete, messy, and had many unanswered questions.	The report was complete, messy, and had many unanswered questions.	The report was complete, neat, and had 1 or 2 unanswered questions.	The report was complete, neat, and all the questions had been answered.	
II. Coding Design and Patterns	The program had inappropriate elements and structures, incorrect indention, and poor program documentation.	The program had corrected indention, but inappropriate elements and structures and poor program documentation.	The program had corrected indention, adequate program documentation, but inappropriate elements and structures.	The program had appropriate elements and structures, correct indention, and excellent program documentation.	
III. Findings, Observations, and Comments	The findings. observations, and comments were not based on the gathered data and results. All thoughts were inconsistent and unclear.	The findings. observations, and comments based on the gathered data and results but were not elaborated. Not all thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were mostly elaborated. Most of the thoughts were consistent and clear.	The findings. observations, and comments were based on the gathered data and results and were completely elaborated. All the thoughts were consistent and clear.	
IV. Grammar and Composition	The words used were inappropriate, wrong grammar usage, and poor sentence construction was observed.	The words used were appropriate; however, bad grammar and poor sentence construction were observed.	The words used were appropriate, had proper grammar usage, but poor sentence construction was observed.	The words used were appropriate, had proper grammar usage, and excellent sentence construction.	
SCORE:					

CHECKED			
Rating		**Signature**	
		Date	